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$$P_1\left[M_1^{\,a}L_1^{\,b}T_1^{\,c}\right]=P_2\left[M_2^{\,a}L_2^{\,b}T_2^{\,c}\right]$$

We have $P_2=P_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$

$M_1 = 1g, M_2 = 1kg$

$L_1 = 1\ cm, L_2 = 1m$

$T_1 = 1\ s, T_2 = 1s$

So a =1, b= -1, and c = -2

Then

$$P_2=76\times13.6\times980\left[\frac{1g}{1kg}\right]^1\left[\frac{1cm}{1m}\right]^{-1}\left[\frac{1s}{1s}\right]^{-2}$$

$$=76\times13.6\times980\left[\frac{10^{-3}kg}{1kg}\right]^1\left[\frac{10^{-2}m}{1m}\right]^{-1}\left[\frac{1s}{1s}\right]^{-2}$$

$$=76\times13.6\times980\times[10^{-3}]\times10^{2}$$

$$P_2=1.01\times10^{5}\ Nm^{-2}$$

EXAMPLE 1.13

If the value of universal gravitational constant in SI is $6.6\times10^{-11}\ Nm^2\ kg^{-2}$, then find its value in CGS System?

Solution

Let G_{SI} be the gravitational constant in the SI system and G_{cgs} in the cgs system. Then

$$G_{SI}=6.6\times10^{-11}Nm^2kg^{-2}$$

$$G_{cgs}=?$$

$$n_2=n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$

$$G_{cgs}=G_{SI}\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$

$M_1 = 1\ kg \quad L_1 = 1m \quad T_1 = 1s$

$M_2 = 1\ g \quad L_2 = 1\ cm \quad T_2 = 1s$

The dimensional formula for G is $M^{-1}L^3T^{-2}$

a = -1 b = 3 and c = -2

$$G_{cgs}=6.6\times10^{-11}\left[\frac{1kg}{1g}\right]^{-1}\left[\frac{1m}{1cm}\right]^3\left[\frac{1s}{1s}\right]^{-2}$$

$$=6.6\times10^{-11}\left[\frac{1kg}{10^{-3}kg}\right]^{-1}\left[\frac{1m}{10^{-2}m}\right]^3\left[\frac{1s}{1s}\right]^{-2}$$

$$=6.6\times10^{-11}\times10^{-3}\text{x}\ 10^{6}\times1$$

$$G_{cgs}=6.6\times10^{-8}\ dyne\ cm^2\ g^{-2}$$

(ii) To check the dimensional correctness of a given physical equation

Let us take the equation of motion $v = u + at$

Apply dimensional formula on both sides

$[LT^{-1}] = [LT^{-1}] + [LT^{-2}]\,[T]$

$$[LT^{-1}] = [LT^{-1}] + [LT^{-1}]$$

(Quantities of same dimension only can be added)

We see that the dimensions of both sides are same. Hence the equation is dimensionally correct.

EXAMPLE 1.14

Check the correctness of the equation $\frac{1}{2}mv^2=mgh$ using dimensional analysis method.